

1 Introduction

Oblivious Transfer (OT) is a fundamental cryptographic primitive that enables two parties to transfer information while preserving the privacy of their inputs. In a 1-out-of-2 Oblivious Transfer (1-out-of-2 OT) protocol, a sender holds two messages, and a receiver obtains exactly one message of its choice. The receiver's choice remains hidden from the sender, while the unselected message remains hidden from the receiver. OT is a fundamental building block in secure multiparty computation (MPC), particularly in protocols based on Yao's garbled circuits.

1.1 Scheme Definition

A 1-out-of-2 Oblivious Transfer scheme is an interactive protocol between a sender S and a receiver R , parameterized by a security parameter λ . It is defined by the tuple of probabilistic polynomial-time algorithms

$$\Pi_{\text{OT}} = (\text{Setup}, \text{Send}, \text{Receive}),$$

where the algorithms operate as follows:

- $\text{Setup}(1^\lambda)$: Generates the public parameters pp required by the protocol.
- $\text{Send}(\text{pp}, m_0, m_1)$: The sender takes two messages $m_0, m_1 \in \{0, 1\}^\ell$ as input and participates in the interactive protocol.
- $\text{Receive}(\text{pp}, b)$: The receiver takes a choice bit $b \in \{0, 1\}$ as input and participates in the interactive protocol to obtain the selected message m_b .

The scheme must satisfy the following properties.

Correctness. For every pair of messages $m_0, m_1 \in \{0, 1\}^\ell$ and every choice bit $b \in \{0, 1\}$, honestly executing parties satisfy

$$\Pr[\text{out}_R = m_b] \geq 1 - \text{negl}(\lambda),$$

where out_R denotes the receiver's output and the probability is taken over the randomness of the protocol.

Sender Privacy. A malicious receiver should not learn information about the unselected message m_{1-b} beyond what can be inferred from the selected message m_b . Formally, for any probabilistic polynomial-time receiver adversary $\mathcal{A}$, its views in executions with message pairs (m_0, m_1) and (m'_0, m'_1) must be computationally indistinguishable whenever the messages selected by its choice bit are identical.

Receiver Privacy. A malicious sender should not learn the receiver's choice bit. Formally, for any probabilistic polynomial-time sender adversary $\mathcal{A}$, its views in executions with receiver choices $b = 0$ and $b = 1$ must be computationally indistinguishable for the same sender inputs.

Here, $\text{negl}(\lambda)$ denotes a negligible function in the security parameter λ , and computational indistinguishability is denoted by $\approx_c$.

1.2 1-out-of- n OT

The definition can be generalized to 1-out-of- n Oblivious Transfer (1-out-of- n OT), where the sender holds n messages $(m_1, \dots, m_n)$ and the receiver holds a selection index $b \in \{1, \dots, n\}$. The receiver obtains only the selected message m_b , while the sender learns nothing about the selection index b and the receiver learns no information about the remaining messages $\{m_i\}_{i \neq b}$. The correctness, sender privacy, and receiver privacy requirements are defined analogously to those of 1-out-of-2 OT.

2 Example Constructions

2.1 Diffie–Hellman-Based Oblivious Transfer

Here, we present the 1-out-of-2 OT protocol proposed by Chou and Orlandi [1].

Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of prime order p . Let H be a hash function used for key derivation, and let (Enc, Dec) be a symmetric encryption scheme. The sender S holds messages (M_0, M_1) , and the receiver R holds a choice bit $c \in \{0, 1\}$.

Protocol.

1. **Sender:** Choose $a \xleftarrow{\$} \mathbb{Z}_p$ and compute $A = g^a$. Send A to R .
2. **Receiver:** Choose $b \xleftarrow{\$} \mathbb{Z}_p$ and compute

$$B = \begin{cases} g^b, & c = 0, \\ Ag^b, & c = 1. \end{cases}$$

Send B to S .

3. **Sender:** Compute

$$k_0 = H(B^a), \quad k_1 = H((B/A)^a),$$

and encrypt the messages:

$$e_0 = \text{Enc}_{k_0}(M_0), \quad e_1 = \text{Enc}_{k_1}(M_1).$$

Send (e_0, e_1) to R .

4. **Receiver:** Compute $k_R = H(A^b)$ and output

$$M_c = \text{Dec}_{k_R}(e_c).$$

Correctness. If $c = 0$, then $B = g^b$, and hence $B^a = g^{ab} = A^b$. Thus $k_0 = k_R$. If $c = 1$, then $B = Ag^b$, so $B/A = g^b$ and $(B/A)^a = g^{ab} = A^b$. Thus $k_1 = k_R$. In either case, the receiver derives the encryption key corresponding to its choice and correctly recovers M_c by decryption.

Sender Privacy (Intuition). The receiver knows $A = g^a$, B , and b , and can compute the selected key from $A^b = g^{ab}$. Recovering the unselected message requires the corresponding unselected key, which depends on the additional group element g^{a^2} . Computing this element from g and $A = g^a$ is an instance of the Computational Diffie–Hellman problem. Under the relevant computational hardness assumption and appropriate encryption and hash-function security, the unselected message remains hidden.

Receiver Privacy. For either choice bit, B is uniformly distributed in $\mathbb{G}$ because b is uniformly random. Therefore, the distribution of B does not reveal c to the sender.

2.2 Endemic OT: A post-quantum 1-2 OT protocol

Here, we present the 1-out-of-2 OT protocol proposed by Masny and Rindal [2]. This is a post-quantum oblivious transfer protocol.

Let $\text{UKA} = (A, B, \text{Key})$ be a two-message uniform key agreement protocol over a group $\mathbb{G}$ with operation $\oplus$. Let H_0, H_1 be independent random oracles, where $H_i : \mathbb{G} \rightarrow \mathbb{G}$. The sender S holds no fixed messages; the protocol generates OT strings s_0, s_1 , and the receiver holds a choice bit $c \in \{0, 1\}$.

Protocol.

1. **Receiver:** Choose $r_{1-c} \xleftarrow{\$} \mathbb{G}$ and run $(t_A, m_A) \leftarrow A()$. Compute

$$r_c = m_A \oplus H_c(r_{1-c}).$$

Send (r_0, r_1) to S .

2. **Sender:** For each $i \in \{0, 1\}$, compute

$$m_{A,i} = r_i \oplus H_i(r_{1-i}),$$

run $(t_{B,i}, m_{B,i}) \leftarrow B(m_{A,i})$, and compute

$$s_i = \text{Key}(t_{B,i}, m_{A,i}).$$

Send $(m_{B,0}, m_{B,1})$ to R and output (s_0, s_1) .

3. **Receiver:** Compute

$$s_c = \text{Key}(t_A, m_{B,c})$$

and output (c, s_c) .

Correctness. By construction, $r_c = m_A \oplus H_c(r_{1-c})$, so

$$m_{A,c} = r_c \oplus H_c(r_{1-c}) = m_A.$$

Thus, the receiver and sender use the same key agreement input for index c . By correctness of UKA,

$$s_c = \text{Key}(t_A, m_{B,c}) = \text{Key}(t_{B,c}, m_{A,c}),$$

except with negligible probability. Therefore, the receiver obtains the selected OT string.

Receiver Privacy. The sender receives (r_0, r_1) . For an honest receiver, the unselected value r_{1-c} is uniform, while the selected value is masked using the random oracle. Under the random-oracle model and the uniformity of the key agreement message, the sender's view does not reveal the choice bit, up to the security error of the underlying construction.

Sender Privacy. The receiver obtains only the key-agreement output corresponding to its selected index. The unselected OT string is protected by the key-indistinguishability of the underlying uniform key agreement protocol. The formal security argument in the paper uses programmable random oracles and a simulation-based proof, with rewinding of a malicious receiver. The resulting construction achieves endemic security in the random-oracle model, assuming the underlying key agreement satisfies the required uniformity and key-indistinguishability properties.

References

- [1] T. Chou and C. Orlandi, "The Simplest Protocol for Oblivious Transfer," in *Progress in Cryptology – LATINCRYPT 2015*, Lecture Notes in Computer Science, vol. 9230, pp. 40–58, Springer, 2015.
- [2] D. Masny and P. Rindal, "Endemic Oblivious Transfer," Cryptology ePrint Archive, Report 2019/706, 2019.

- [3] S. Dong, H. Cui, K. Zhang, K. Yang, and Y. Yu, “A Simple Post-Quantum Oblivious Transfer Protocol from Mod-LWR,” Cryptology ePrint Archive, Paper 2024/1116, 2024. <https://eprint.iacr.org/2024/1116>